

LINEAR ALGEBRA & MULTIVARIABLE CALCULUS: 100 PRACTICE PROBLEMS

1. Let $\mathbf{u} = \langle 2, -1, 5 \rangle$ and $\mathbf{v} = \langle -1, 3, 4 \rangle$. Find a vector $\mathbf{w}$ of magnitude 6 that is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$.
2. Determine if the four points $A(1, 0, 2)$, $B(2, 3, -1)$, $C(3, -1, 5)$, and $D(5, 2, 2)$ lie in the same plane. If they do, find the equation of the plane. If not, find the volume of the tetrahedron they define.
3. Find the acute angle between the main diagonal of a cube and one of its adjacent edges.
4. Let the line L_1 be given by the symmetric equations $\frac{x-1}{2} = \frac{y+2}{-1} = \frac{z-3}{3}$ and the line L_2 be given by $x = 2 + t$, $y = 3 - 2t$, $z = -1 + 4t$. Determine whether the lines are parallel, intersecting, or skew. If they intersect, find the point of intersection. If they are skew, find the shortest distance between them.
5. Find the equation of the plane that contains the point $P(2, -1, 3)$ and the line of intersection of the planes $2x - y + z = 5$ and $x + 2y - z = 1$.

6. A particle is moved along the line segment from $P(1, 2, -1)$ to $Q(3, 8, 5)$ by a constant force $\mathbf{F} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$. Find the work done by the force.

7. Find the distance from the point $S(1, 1, 5)$ to the line through $P(0, 1, 2)$ and $Q(2, 3, 0)$.

8. Solve the system of linear equations using Gaussian elimination: [

$$\begin{cases} 2x_1 - 3x_2 + 4x_3 - x_4 = 3 \\ x_1 + 2x_2 - x_3 + 2x_4 = -1 \\ 3x_1 - x_2 + 2x_3 + x_4 = 4 \\ -x_1 + 5x_2 - 5x_3 + 3x_4 = -5 \end{cases}$$

]

9. For what values of k does the following system have no solution, a unique solution, or infinitely many solutions? [

$$\begin{cases} x + y + kz = 1 \\ x + ky + z = 1 \\ kx + y + z = 1 \end{cases}$$

]

10. Find the inverse of the matrix [$A =$

$$\begin{pmatrix} 1 & 2 & -1 \\ 2 & 5 & 1 \\ -1 & -1 & 4 \end{pmatrix}$$

] using elementary row operations.

11. Compute the determinant of the matrix [B =

$$\begin{pmatrix} 2 & -1 & 3 & 0 \\ 1 & 0 & 2 & -1 \\ -3 & 2 & 0 & 4 \\ 0 & 1 & -1 & 3 \end{pmatrix}$$

] by using cofactor expansion along the second column.

12. Find the area of the triangle in $\mathbb{R}^3$ with vertices $P(1, -1, 2)$, $Q(2, 0, 5)$, and $R(3, 2, -1)$.

13. Find the projection of the vector $\mathbf{b} = \langle 1, 2, 3 \rangle$ onto the column space of the matrix [A =

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}$$

]

14. Use Cramer's rule to solve the system: [

$$\begin{cases} 3x_1 - x_2 + 2x_3 = 6 \\ -x_1 + 2x_2 - x_3 = -2 \\ 2x_1 - x_2 + x_3 = 3 \end{cases}$$

]

15. Find the transition matrix from the basis $\mathcal{B} = \{\langle 1, 1 \rangle, \langle 1, -1 \rangle\}$ to the basis $\mathcal{C} = \{\langle 2, 3 \rangle, \langle 1, 2 \rangle\}$ in $\mathbb{R}^2$.

16. Find the distance between the parallel planes $3x - y + 2z = 6$ and

$$6x - 2y + 4z = -4.$$

17. Find the equation of the plane containing the line $x = 3 + 2t$, $y = t$, $z = 8 - t$ and parallel to the line $x = 5 - t$, $y = 2 + 3t$, $z = 1 + 2t$.

18. Traffic flow through a network of four intersections is described by the system:
[

$$\begin{cases} x_1 - x_2 = 100 \\ x_2 - x_3 = -50 \\ x_3 - x_4 = 150 \\ x_4 - x_1 = -200 \end{cases}$$

] Determine the general flow pattern. If the flow x_4 must be at least 50 vehicles per hour, what are the minimum flows on the other streets?

19. Four grid points on a metal plate under thermal equilibrium have temperatures T_1 , T_2 , T_3 , and T_4 . According to the Mean-Value Property, each temperature is the average of its four closest neighbors. The boundary temperatures are:

- Neighbors of T_1 are T_2 , T_3 , 30°C , and 10°C .
- Neighbors of T_2 are T_1 , T_4 , 40°C , and 20°C .
- Neighbors of T_3 are T_1 , T_4 , 10°C , and 30°C .

- Neighbors of T_4 are T_2 , T_3 , 40°C , and 30°C . Set up and solve a system of linear equations to find the equilibrium temperatures T_1 , T_2 , T_3 , and T_4 .

20. Balance the chemical equation for the combustion of propane: $[x_1 \text{C}_3\text{H}_8 + x_2 \text{O}_2 \rightarrow x_3 \text{CO}_2 + x_4 \text{H}_2\text{O}]$ by setting up and solving a homogeneous system of linear equations. Find the smallest positive integer coefficients.

21. Determine the currents I_1 , I_2 , and I_3 in a three-loop electrical network governed by the equations: [

$$\begin{cases} 12I_1 - 3I_2 = 30 \\ -3I_1 + 6I_2 - I_3 = 5 \\ -I_2 + 3I_3 = -25 \end{cases}$$

]

22. Let $A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 2 & -3 & 5 \end{pmatrix}$. Find the matrix $A^3 - 7A^2 + 5A - 3I$.

23. Find the equation of the plane that passes through the points $P(1, 3, 2)$ and $Q(3, -1, 6)$ and is perpendicular to the plane $x - 2y + z = 5$.

24. Find the parametric equations of the line that passes through the point $(1, 0, -3)$ and is parallel to the line of intersection of the planes $x + 2y - z = 2$ and $3x - y + 2z = 1$.

25. Find the volume of the parallelepiped determined by the vectors $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j}$, $\mathbf{b} = \mathbf{i} + \mathbf{j} - \mathbf{k}$, and $\mathbf{c} = 3\mathbf{i} - \mathbf{k}$.

- 26.** Find the distance from the point $(3, -1, 4)$ to the plane $2x - y + 2z = 5$.
- 27.** Find the coordinates of the point where the line $x = 1 + 2t$, $y = -1 + 3t$, $z = 2 - t$ intersects the plane $3x - y + 2z = 12$.
- 28.** Find the symmetric equations of the line that passes through the point $(2, 1, -1)$ and is perpendicular to the plane containing the points $(1, 0, 0)$, $(0, 2, 0)$, and $(0, 0, 3)$.
- 29.** Find the cosine of the angle between the planes $x + y + z = 1$ and $2x - y - 2z = 5$.
- 30.** Determine whether the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ is parallel to, lies in, or intersects the plane $2x + 3y + 12z = 13$. If it intersects, find the point of intersection.
- 31.** Find the vector equation of the line of intersection of the planes $2x + y - z = 3$ and $x - y + 2z = 1$.
- 32.** Find the equation of the plane that contains the point $(1, 2, 3)$ and is perpendicular to the line $x = 2 + t$, $y = 3 - 2t$, $z = t$.
- 33.** An economy has three sectors: Agriculture, Manufacturing, and Services. The consumption matrix is: $[C =$

$$\begin{pmatrix} 0.5 & 0.4 & 0.2 \\ 0.2 & 0.3 & 0.1 \\ 0.1 & 0.1 & 0.3 \end{pmatrix}$$

] Find the production level required to satisfy a final demand of 50 units of Agriculture, 30 units of Manufacturing, and 20 units of Services.

34. Find the least-squares line $y = \beta_0 + \beta_1 x$ for the data points $(1, 2)$, $(2, 5)$, $(3, 5)$, and $(4, 8)$.

35. Find the distance from the point $(1, -1, 1)$ to the line $\frac{x-2}{2} = \frac{y+1}{-1} = z - 3$.

36. Let $A = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 1 & 3 \\ 1 & 2 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 5 \\ 3 \end{pmatrix}$. Use the LU factorization of A to solve $A\mathbf{x} = \mathbf{b}$.

37. For the matrix $A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & -4 \\ -1 & -3 & 2 \end{pmatrix}$, find a basis for the null space and a basis for the column space.

38. Let $\mathbf{u} = \langle 1, 2, -1 \rangle$ and $\mathbf{v} = \langle 3, 0, 2 \rangle$. Find a vector $\mathbf{w}$ in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$ that is perpendicular to the vector $\mathbf{a} = \langle 1, 1, 1 \rangle$.

39. Find the equation of the plane that passes through the point $(1, 1, 1)$ and contains the line $x = 2t$, $y = 1 + t$, $z = -t$.

40. Compute the distance between the skew lines $L_1: x = 1 + t$, $y = 2 - t$, $z = 3 + 2t$ and $L_2: x = 2 - s$, $y = 1 + 2s$, $z = 4 - s$.

41. Find the values of a and b for which the system of equations: [

$$\begin{cases} x + y + z = 2 \\ 2x + 3y + 2z = 5 \\ 2x + 3y + (a^2 - 1)z = a + 1 \end{cases}$$

] has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions.

42. Let $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. Find a matrix P that diagonalizes A orthogonally, and compute the diagonal matrix $D = P^T A P$.

43. Find the equation of the plane that is parallel to the plane $x - 2y + 2z = 3$ and is at a distance of 4 units from it.

44. Find the parametric equations of the line that is the projection of the line $x = t$, $y = 1 + t$, $z = 2t$ onto the plane $x + y + z = 1$.

45. Find the volume of the tetrahedron with vertices $(0, 0, 0)$, $(1, 1, 1)$, $(2, 3, 1)$, and $(1, 2, 3)$.

46. Let $\mathbf{u}$ and $\mathbf{v}$ be vectors such that $\|\mathbf{u}\| = 3$, $\|\mathbf{v}\| = 5$, and $\|\mathbf{u} + \mathbf{v}\| = 7$. Find the dot product $\mathbf{u} \cdot \mathbf{v}$ and the angle between them.

47. Compute the inverse of the matrix: [$A =$

$$\begin{pmatrix} 1 & 0 & 3 \\ 2 & -1 & 5 \\ 0 & 1 & 2 \end{pmatrix}$$

] using the adjugate matrix formula.

48. Evaluate the determinant of the $n \times n$ matrix: [A =

$$\begin{pmatrix} 2 & 1 & 0 & \dots & 0 \\ 1 & 2 & 1 & \dots & 0 \\ 0 & 1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$

] for $n = 4$.

49. Find the equation of the plane that passes through the line of intersection of the planes $x - z = 1$ and $y + 2z = 3$ and is perpendicular to the plane $x + y - 2z = 1$.

50. Let $P(1, 2, -1)$ be a point and L be the line $x = 3 - t$, $y = 2 + 2t$, $z = 1 + t$. Find the coordinates of the point Q on L that is closest to P .

51. Find the symmetric equations of the line passing through the point $(1, -2, 3)$ and perpendicular to both lines $L_1: x = 2t$, $y = -t$, $z = 3t$ and $L_2: x = 1 + s$, $y = 2 - 2s$, $z = -1 + s$.

52. Find the equation of the plane that contains the point $(1, 2, -3)$ and is parallel to the vectors $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 1, 1, 1 \rangle$.

53. Solve the system of equations by using the inverse of the coefficient matrix: [

$$\begin{cases} x + y + z = 6 \\ 2x - y + z = 3 \\ x + y - 2z = 0 \end{cases}$$

]

54. Find the values of x for which the matrix $A = \begin{pmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{pmatrix}$ is singular.
55. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Find a matrix B such that $AB - BA = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, or show that no such matrix exists.
56. Find the area of the parallelogram determined by the vectors $\mathbf{u} = \langle 3, 1, -2 \rangle$ and $\mathbf{v} = \langle 1, 2, 4 \rangle$.
57. Determine if the vectors $\mathbf{v}_1 = \langle 1, 1, 2 \rangle$, $\mathbf{v}_2 = \langle 1, 0, 1 \rangle$, and $\mathbf{v}_3 = \langle 2, 1, 3 \rangle$ are linearly independent. If not, write one of them as a linear combination of the other two.
58. Find the standard matrix of the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that reflects each vector across the yz -plane.
59. Find the standard matrix of the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that rotates vectors counterclockwise by 45° , followed by a projection onto the x -axis.
60. Find the equation of the plane containing the point $(2, 0, 1)$ and perpendicular to the planes $2x + y - z = 3$ and $x - y + z = 1$.
61. Find the distance from the point $(1, 0, 2)$ to the line of intersection of the planes $x + y - z = 2$ and $2x - y + z = 1$.
62. Find the coordinates of the point on the plane $2x - 3y + z = 6$ that is closest to the point $(1, 1, 1)$.

63. Find the angle between the diagonal of a face of a cube and the diagonal of the cube that starts from the same vertex.

64. Find a basis for the row space of the matrix [$A =$

$$\begin{pmatrix} 1 & -2 & 0 & 3 \\ 2 & -3 & -1 & 5 \\ 1 & 0 & -2 & 2 \end{pmatrix}$$

]

65. Let V be the vector space of polynomials of degree at most 2. Find the transition matrix from the basis $\mathcal{B} = \{1, x, x^2\}$ to the basis $\mathcal{C} = \{1, 1 + x, 1 + x + x^2\}$.

66. Find the shortest distance between the lines $x = t, y = 2t, z = 3t$ and $x = s + 1, y = 3s - 1, z = 2s + 2$.

67. Find the equation of the plane containing the line $x = 1 + t, y = 2 - t, z = 3 + 2t$ and the point $(0, 1, 2)$.

68. Find the symmetric equations of the line that is perpendicular to the plane $3x - y + 2z = 4$ and passes through the point of intersection of the line $\frac{x-1}{2} = y = \frac{z+1}{-3}$ and the plane $x + 2y - z = 5$.

69. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Find a formula for A^n where n is a positive integer.

70. Solve the system using LU decomposition: [

$$\begin{cases} 2x + y + z = 5 \\ 4x - 6y = -2 \\ -2x + 7y + 2z = 9 \end{cases}$$

]

71. Let A be a 3×3 matrix such that $\det(A) = 4$. Compute $\det(2A^{-1}A^T(3A)^T)$.

72. Find the equation of the plane containing the parallel lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{-1}$ and $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-3}{-1}$.

73. Let the line L be given by $x = 1 + t$, $y = -1 + 2t$, $z = 3 - t$. Find the equation of the plane containing L that is perpendicular to the plane $2x - y + 3z = 4$.

74. Find the values of a for which the vectors $\mathbf{u} = \langle 1, a, 1 \rangle$, $\mathbf{v} = \langle 2, 1, a \rangle$, and $\mathbf{w} = \langle a, 1, 2 \rangle$ are coplanar.

75. Let $P(1, 0, 1)$ and $Q(2, -1, 3)$. Find the coordinates of the point R on the line segment PQ such that $PR:RQ = 2:3$.

76. Let $\mathbf{u} = \langle 2, 1, -3 \rangle$ and $\mathbf{v} = \langle 1, -2, 1 \rangle$. Write $\mathbf{u}$ as the sum of a vector parallel to $\mathbf{v}$ and a vector orthogonal to $\mathbf{v}$.

77. Find the equation of the plane that contains the point $(1, 3, -2)$ and the line $x = t$, $y = 2t$, $z = 3t$.

- 78.** Determine the value of k such that the line $\frac{x-1}{k} = \frac{y+2}{3} = \frac{z}{2}$ is parallel to the plane $2x - y + 3z = 5$.
- 79.** Find the acute angle between the line $\frac{x-1}{2} = y - 3 = \frac{z+2}{-1}$ and the plane $x + 2y - z = 4$.
- 80.** Find the equation of the plane containing the line $x = 1 + 2t, y = 3 - t, z = 2 + t$ and the point $(2, 0, -1)$.
- 81.** Find the distance from the point $(1, -2, 4)$ to the plane containing the lines $x = t, y = 2t, z = 3t$ and $x = 1 - s, y = 2 + s, z = 3 - s$.
- 82.** Balance the chemical equation for the reaction of sodium bicarbonate and citric acid: $[x_1 \text{ } _3 + x_2 \text{ } _3\text{ } _6\text{ } _5\text{ } _7 + x_3 \text{ } _3\text{ } _6\text{ } _5\text{ } _7 + x_4 \text{ } _2 + x_5 \text{ } _2]$ by solving the corresponding homogeneous linear system for the smallest positive integers.
- 83.** A business analyst finds that the quarterly sales of a product are related to marketing spend x_1 and research spend x_2 by the linear equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. Given the data points (x_1, x_2, y) of $(1, 2, 5)$, $(2, 1, 6)$, $(3, 2, 8)$, and $(2, 3, 7)$, construct the design matrix X and observation vector $\mathbf{y}$, then solve the normal equations to find β_0, β_1 , and β_2 .
- 84.** Let $\mathbf{u} = \langle 1, 1, 0 \rangle$, $\mathbf{v} = \langle 1, 0, 1 \rangle$, and $\mathbf{w} = \langle 0, 1, 1 \rangle$. Express the vector $\mathbf{a} = \langle 2, 3, 4 \rangle$ as a linear combination of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$.

- 85.** Let $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix}$. Find a 2×2 matrix B such that $AB = I$, or show that no such matrix exists.
- 86.** Find the equation of the plane containing the point $(2, -1, 3)$ and parallel to the plane containing the points $(1, 1, 1)$, $(2, 0, -1)$, and $(0, 2, 3)$.
- 87.** Find the symmetric equations of the line that is the intersection of the planes $3x + y - 2z = 5$ and $2x - y + z = 2$.
- 88.** Compute the volume of the tetrahedron with vertices at the origin and the points of intersection of the plane $3x + 4y + 6z = 24$ with the coordinate axes.
- 89.** Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$. Find the determinant of A using the properties of determinants and row reduction.
- 90.** Use the least-squares method to fit a parabola $y = \beta_0 + \beta_1 x + \beta_2 x^2$ to the data points $(-1, 2)$, $(0, 1)$, $(1, 2)$, and $(2, 5)$.
- 91.** Find the distance from the point $(2, -1, 3)$ to the line through $(1, 0, 1)$ parallel to the vector $\langle 2, -1, 1 \rangle$.
- 92.** Let $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}$. Find the eigenvalues and eigenvectors of the product matrix AB .

- 93.** Find the equation of the plane containing the point $(1, 1, 1)$ and the line of intersection of the planes $x - y + z = 2$ and $2x + y - z = 3$.
- 94.** Find the shortest distance between the line $x = 1 + 2t, y = -1 + t, z = 3t$ and the line $x = 2 - s, y = 1 + 3s, z = -1 + s$.
- 95.** Find the standard matrix of the linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that projects vectors orthogonally onto the plane $x - 2y + z = 0$.
- 96.** Solve the network flow problem where the nodes satisfy the flow conservation equations: [

$$\begin{cases} x_1 - x_2 + x_3 = 50 \\ x_2 - x_4 = 30 \\ x_4 - x_3 - x_5 = -20 \\ x_5 - x_1 = -60 \end{cases}$$

] Determine the general solution and find the flow values if all flows must be non-negative.

- 97.** Find the coordinates of the point on the line $x = 2 - t, y = 1 + 2t, z = 3 - 2t$ that is closest to the point $(1, -1, 2)$.
- 98.** Find the equation of the plane containing the points $(1, -1, 2)$ and $(2, 0, 3)$ and perpendicular to the plane $3x - y + 2z = 5$.
- 99.** Determine the value of k for which the line $x = 1 + kt, y = -1 + 3t, z = 2t$ is parallel to the line $x = 2 - 2s, y = 3 + s, z = 1 - s$.

00. Find the equation of the sphere with its center on the plane $2x - y + z = 3$ and passing through the three points $(1, 2, -1)$, $(2, 1, 3)$, and $(-1, 0, 1)$.